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RISC-V Based In-Memory Computing Architecture for Energy Efficient MAC Operations

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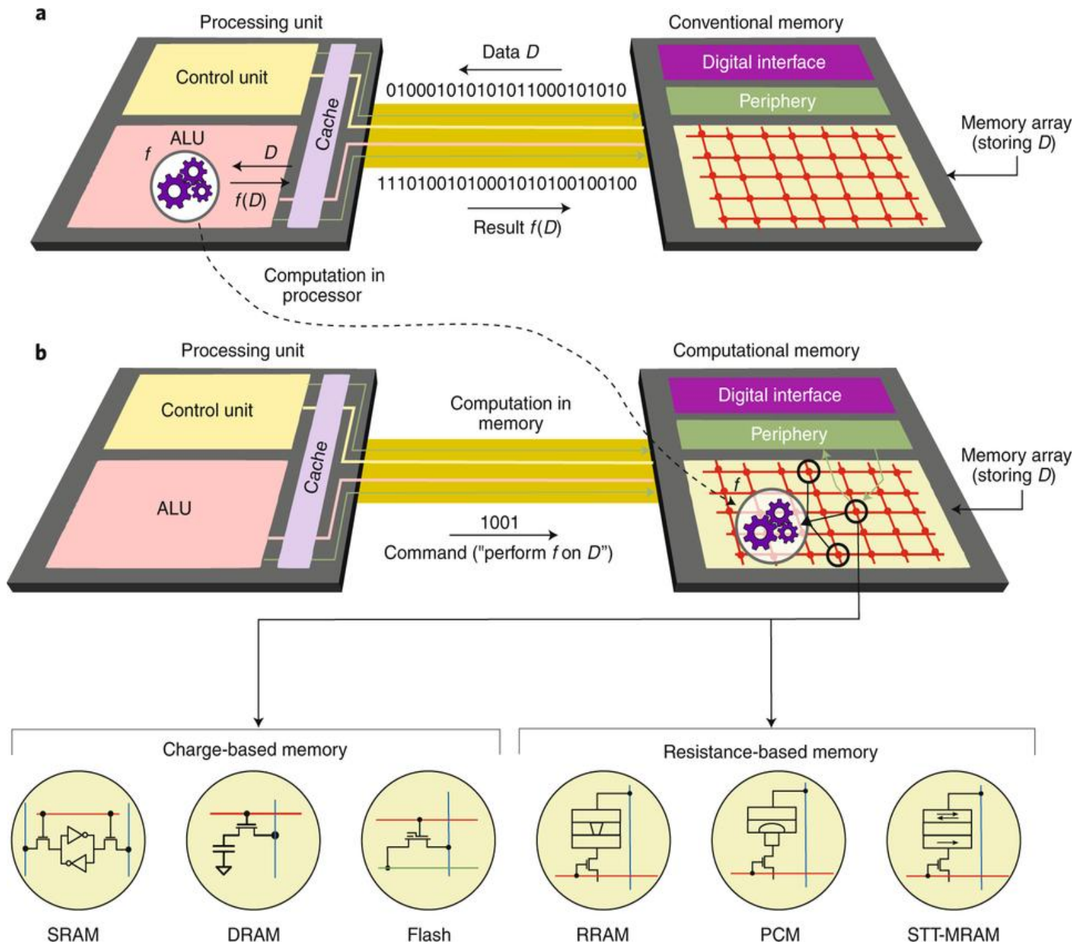
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PROBLEM STATEMENT

- Conventional computing architectures require frequent data transfer between memory and the processor, resulting in significant **energy consumption and performance bottlenecks.**
- MAC intensive applications require repeated memory accesses and computations, which **increase execution latency and energy consumption**.
- The separation of memory and computation in conventional RISC-V architectures creates a **data movement bottleneck** that limits the efficiency of data intensive workloads.
- Therefore, an **SRAM based Compute-in-Memory** architecture integrated with a **RISC-V processor** is required to perform MAC operations closer to the stored data and **reduce unnecessary data movement**



OBJECTIVES

1. Design and implement a **RISC-V based architecture integrated with an SRAM based Compute in Memory (CIM)** unit for efficient Multiply Accumulate (MAC) operations.	**2.** Develop a **custom instruction and control mechanism to trigger MAC computations** directly within the memory subsystem, reducing unnecessary processor memory data movement.
3. Implement and synthesize both **conventional memory plus MAC and proposed SRAM-CIM** architectures using Cadence Genus for a fair comparative analysis.	**4.** Evaluate and compare **area, power, delay, energy per MAC, memory accesses and throughput** to assess the efficiency of the proposed architecture.

LITERATURE REVIEW

S.NO	TITLE OF THE PAPER	JOURNAL	AUTHOR(s)	INFERENCE
[1]	AI-PiM—Extending the RISC-V Processor with Processing-in-Memory Functional Units for AI Inference at the Edge of IoT	Frontiers in Electronics, 2022	Vaibhav Verma, Mircea R. Stan	Integrates PiM functional units directly into the RISC-V pipeline using custom instructions, reducing data movement and improving AI inference performance.
[2]	RDCIM: RISC-V Supported Full-Digital Computing-in-Memory Processor With High Energy Efficiency and Low Area Overhead	IEEE Transactions on Circuits and Systems I, 2024	W. Yi et al.	Demonstrates a fabricated 55-nm RISC-V digital CIM processor for efficient MAC operations, using custom RISC-V instructions and an adaptive accumulator to improve energy and area efficiency.

LITERATURE REVIEW

S.NO	TITLE OF THE PAPER	JOURNAL	AUTHOR(s)	INFERENCE
[3]	PipeCIM: A High-Throughput Computing-In-Memory Microprocessor with Nested Pipeline and RISC-V Extended Instructions	IEEE Transactions on Circuits and Systems I, 2024	T. Chen et al.	Uses a 55-nm CIM prototype with RISC-V extended instructions and hierarchical pipelining to improve MAC utilization, throughput, and energy efficiency.
[4]	CIMR-V: An End-to-End SRAM-Based CIM Accelerator with RISC-V for AI Edge Device	IEEE/Related Conference Publication, 2025	Y.-C. Guo et al.	Combines SRAM-based CIM with RISC-V and CIM-specific instructions to reduce data movement and enable end-to-end AI inference; implemented in 28-nm technology.

LITERATURE REVIEW

S.NO	TITLE OF THE PAPER	JOURNAL	AUTHOR(s)	INFERENCE
[5]	In-Pipeline Integration of Digital In-Memory-Computing into RISC-V Vector Architecture to Accelerate Deep Learning	arXiv, 2026	T. Spagnolo et al.,	Tightly couples a digital IMC unit with the RISC-V execution pipeline using custom instructions, demonstrating significant acceleration for deep-learning workloads.
[6]	Deep Learning on RISC-V Platforms at the Edge: A Perspective on the Hardware and Software Support	ACM Computing Surveys	Survey	Reviews RISC-V-based AI accelerators and IMC approaches, highlighting the importance of hardware–software integration, custom instructions, and efficient memory access for edge AI.

RESEARCH GAP IDENTITFIED

• **Limited lightweight RISC-V IMC integration:** Existing architectures often use complex or highly specialized IMC designs, a simpler tightly coupled RISC-V IMC architecture for MAC intensive workloads is still worth exploring.	• **High hardware implementation overhead:** Existing IMC approaches may introduce significant area, power, and control complexity, creating a need for a more hardware efficient architecture.
• **Limited complete ASIC evaluation:** Many works emphasize architectural or simulation level results, with less focus on a complete RTL to GDSII implementation, including physical design, timing, power, DRC and LVS.	• **Lack of direct baseline comparison:** A systematic comparison of conventional RISC-V processor memory architecture vs. RISC-V IMC architecture across data movement, latency, power, energy and area is needed.

PROPOSED WORK

1. **Design of RISC-V Processor**
   - Implement a lightweight RISC-V processor core.
   - Support the required arithmetic, load/store, and control instructions.
   - Integrate a custom instruction for triggering the IMU operation.

2. **Design of Conventional MAC Baseline**
   - Implement the conventional architecture where MAC operations are performed within/through the RISC-V datapath.
   - Establish a baseline for comparison in terms of:
     - Area
     - Power
     - Delay
     - Energy per MAC
     - Data movement

3. Design of SRAM-Based CIM / IMU

- Develop an SRAM-based In-Memory Unit (IMU).
- Store operands/weights in the SRAM array.
- Perform multiplication and accumulation close to/in the memory array.
- Minimize data movement between memory and processor.

4. Custom RISC-V Instruction

- Introduce an instruction such as:

IMU_MAC

- RISC-V issues the instruction.
- The instruction activates/configures the IMU.
- The IMU performs the MAC operation and returns the result/status.

5. IMU Controller

- Design the control logic for:
  - Start
  - Address generation
  - SRAM read/write control
  - MAC sequencing
  - Accumulation
  - Completion/status
- Generate a done signal/interrupt to the RISC-V core.

6. RTL Integration

- Integrate: **RISC-V + SRAM + IMU Controller + CIM/MAC array** into a complete SoC-level RTL architecture.

7. Functional Verification

- Verify the conventional MAC and CIM-MAC architectures using the same input data and operations.
- Confirm that both produce identical MAC results.

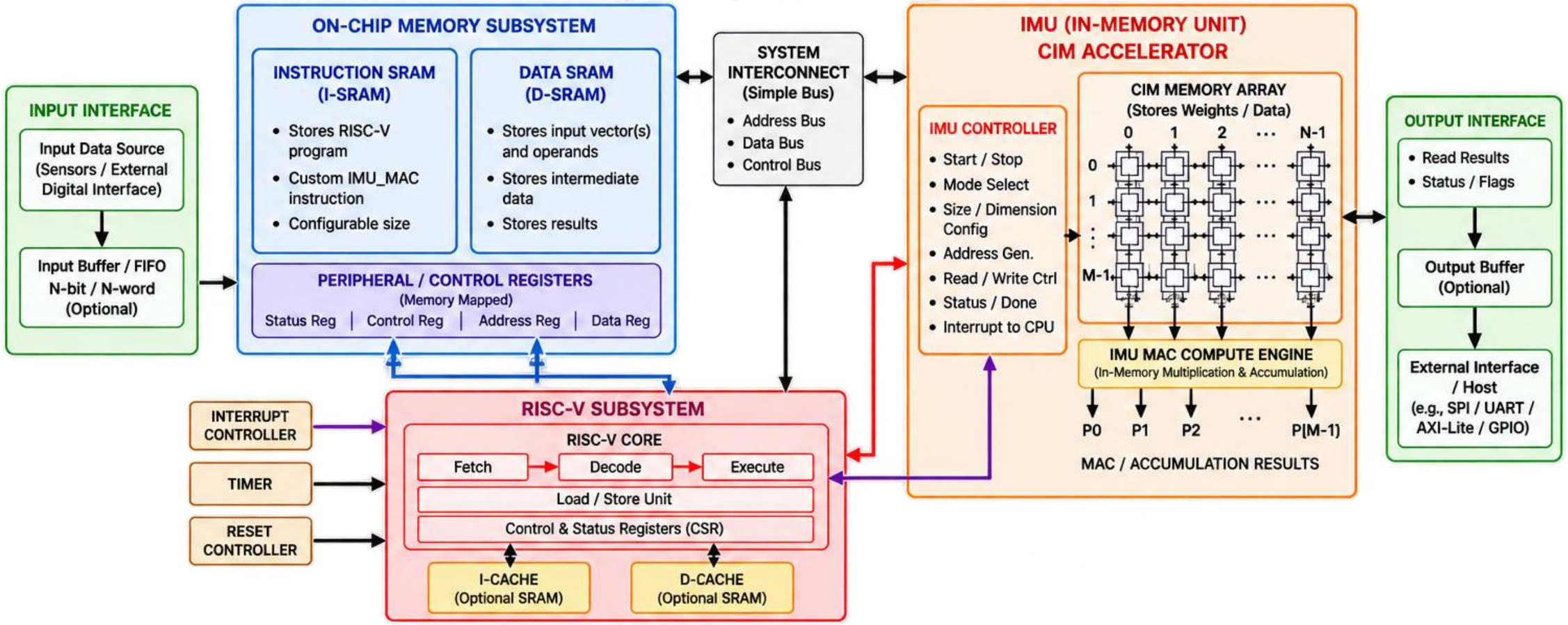
8. Cadence Synthesis and Evaluation

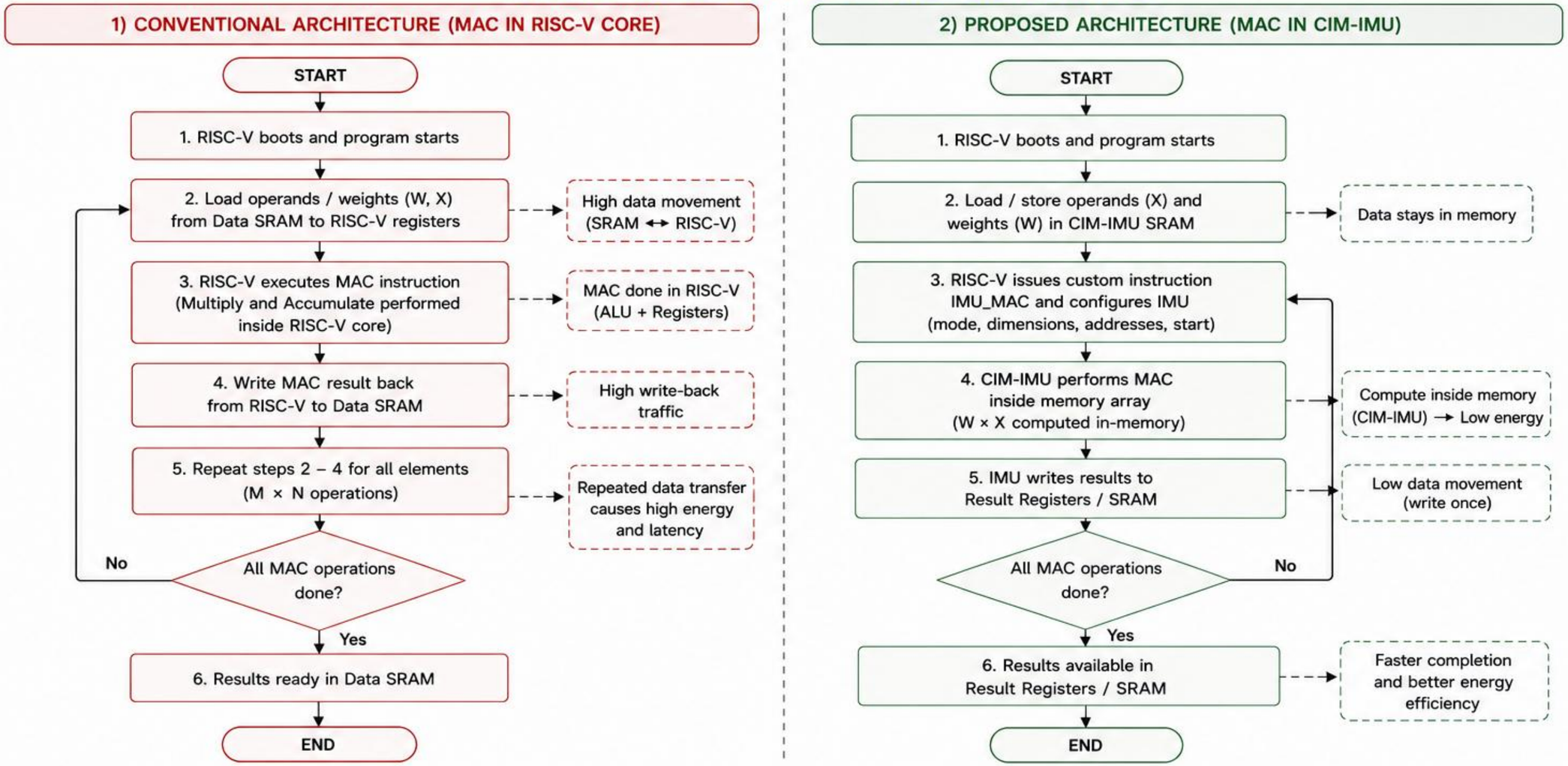
- Synthesize both architectures using Cadence Genus.
- Compare:
  - Area
  - Power
  - Timing/delay
  - Energy per MAC
  - Throughput
  - Memory/data-transfer activity

9. Performance Comparison

- Demonstrate the advantage of moving MAC computation from the conventional RISC-V datapath to the CIM/IMU.
- Analyze the reduction in memory-to-compute data movement and its effect on energy and latency.

BLOCK DIAGRAM




1) CONVENTIONAL ARCHITECTURE (MAC IN RISC-V CORE)
START
1. RISC-V boots and program starts
2. Load operands / weights (W, X) from Data SRAM to RISC-V registers
High data movement (SRAM ↔ RISC-V)
3. RISC-V executes MAC instruction (Multiply and Accumulate performed inside RISC-V core)
MAC done in RISC-V (ALU + Registers)
4. Write MAC result back from RISC-V to Data SRAM
High write-back traffic
5. Repeat steps 2 – 4 for all elements (M × N operations)
Repeated data transfer causes high energy and latency
All MAC operations done?
No
Yes
6. Results ready in Data SRAM
END
2) PROPOSED ARCHITECTURE (MAC IN CIM-IMU)
START
1. RISC-V boots and program starts
2. Load / store operands (X) and weights (W) in CIM-IMU SRAM
Data stays in memory
3. RISC-V issues custom instruction IMU_MAC and configures IMU (mode, dimensions, addresses, start)
4. CIM-IMU performs MAC inside memory array (W × X computed in-memory)
Compute inside memory (CIM-IMU) → Low energy
5. IMU writes results to Result Registers / SRAM
Low data movement (write once)
All MAC operations done?
No
Yes
6. Results available in Result Registers / SRAM
Faster completion and better energy efficiency
END

SGD GOALS ALLIGNED

SDG 7 - Affordable and Clean Energy : Reduces energy consumption by minimizing processor memory data movement during MAC intensive computations.



SDG 9 - Industry, Innovation and Infrastructure : Develops an innovative RISC-V based computing architecture and demonstrates its implementation through a complete ASIC design flow.



SDG 12 - Responsible Consumption and Production : Improves hardware and computational resource utilization by performing MAC operations close to the stored data.



SDG 13 - Climate Action : Promotes energy efficient computing, helping reduce the operational power consumption and associated carbon footprint of AI hardware.



REFERENCES & PUBLICATIONS

[1] V. Verma and M. R. Stan, “AI-PiM—Extending the RISC-V Processor with Processing-in-Memory Functional Units for AI Inference at the Edge of IoT,” Frontiers in Electronics, vol. 3, 2022, Art. no. 898273, doi: 10.3389/felec.2022.898273.

[2] W. Yi, K. Mo, W. Wang, Y. Zhou, Y. Zeng, Z. Yuan, B. Cheng, and B. Pan, “RDCIM: RISC-V Supported Full-Digital Computing-in-Memory Processor With High Energy Efficiency and Low Area Overhead,” IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 71, no. 4, pp. 1719–1732, 2024, doi: 10.1109/TCSI.2024.3350664.

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[4] Y.-C. Guo, T.-S. Chang, C.-S. Lin, B.-C. Chiou, C.-M. Lai, S.-S. Sheu, W.-C. Lo, and S.-C. Chang, “CIMR-V: An End-to-End SRAM-Based CIM Accelerator With RISC-V for AI Edge Device,” arXiv preprint arXiv:2503.22072, 2025.

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Thank you!